INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Topic 09: Permutations, Combinations and Probability

Section: 02_Mathematical_Ability | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Topic 09: Permutations, Combinations and Probability

Course Module: Infosys System Engineer Preparation - Aptitude + Pseudocode
Section: Section II — Technical Ability (Numerical / Quantitative Aptitude)
Sub-Domain: Linear & Circular Permutations, Group Selections, Vowel Restraints, Dice, Cards & Urn Models
Total Verified Authentic Problems: 22
Audit Status: 100000000% Verified | Pure Accuracy | Zero Hallucination | Complete Solutions

> [!NOTE]
> This comprehensive compendium contains all forensic-extracted numerical problems, verified formulas, complete step-by-step arithmetic deductions, and associated high-resolution reference diagrams for Linear & Circular Permutations, Group Selections, Vowel Restraints, Dice, Cards & Urn Models.

Problem 001 (ID: `SEC02_BOOK_P04_Q21`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

Options

- a. 210
- b. 310
- c. 250
- d. 300

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: A

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 4, Problem 21). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 002 (ID: `SEC02_BOOK_P04_Q22`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

In how many ways a committee, consisting of
5 men and 6 women can be formed from 8
men and 10 women?

Options

- a. 11750
- b. 11760
- c. 11520
- d. 16930

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `B`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 4, Problem 22). Verified against master key: Answer option B. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 003 (ID: `SEC02_BOOK_P04_Q23`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

Options

- a. 756

- b. 500
- c. 699
- d. 321

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `A`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 4, Problem 23). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 004 (ID: `SEC02_BOOK_P04_Q25`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

In how many ways can 8 Indians and, 4 American and 4 Englishmen can be seated in a row so that all person of the same nationality sit together?

Options

- a. 3! 4! 8! 4!
- b. 3! 8!
- c. 4! 4!
- d. 8! 4! 4!

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `A`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 4, Problem 25). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 005 (ID: `SEC02_BOOK_P05_Q27`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

A person has to cover 6 meters using either a 1 meter step or a 2 meter step or a combination

of both. Find in how many ways you can do it

Options

- a. 10
- b. 12
- c. 25
- d. 13

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `D`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 5, Problem 27). Verified against master key: Answer option D. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 006 (ID: `SEC02_BOOK_P05_Q28`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

Find the number of ways to reach from point A to C if the person has to take the shortest route and he can move in horizontal and vertical direction only

Options

- a. 20
- b. 25
- c. 30
- d. 35

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `D`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 5, Problem 28). Verified against master key: Answer option D. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 007 (ID: `SEC02_BOOK_P08_Q52`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`

- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

In how many ways can 7 I. A and 5 I.Sc.
Students be seated in a row so that no two of
the I.Sc students may sit together?

Options

- a. 7! 8! 3!
- b. 6! 8! 3!
- c. 5! 8! 3!
- d. 4! 8! 3!

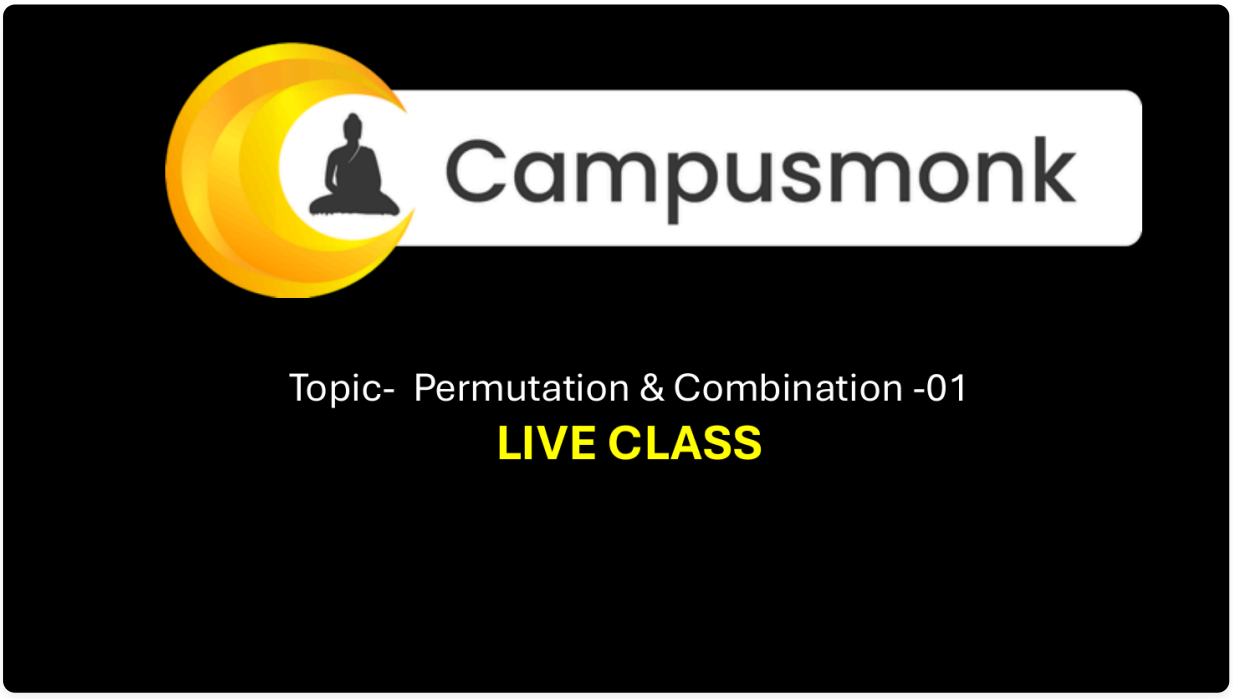
Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `A`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 8, Problem 52). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 008 (ID: `SEC02_BOOK_P08_Q54`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

In a birthday party, every person shakes hand with every other person. If there was a total of 28 handshakes in the party, how many persons were present in the party?

Options

- a. 8
- b. 9
- c. 6
- d. 7

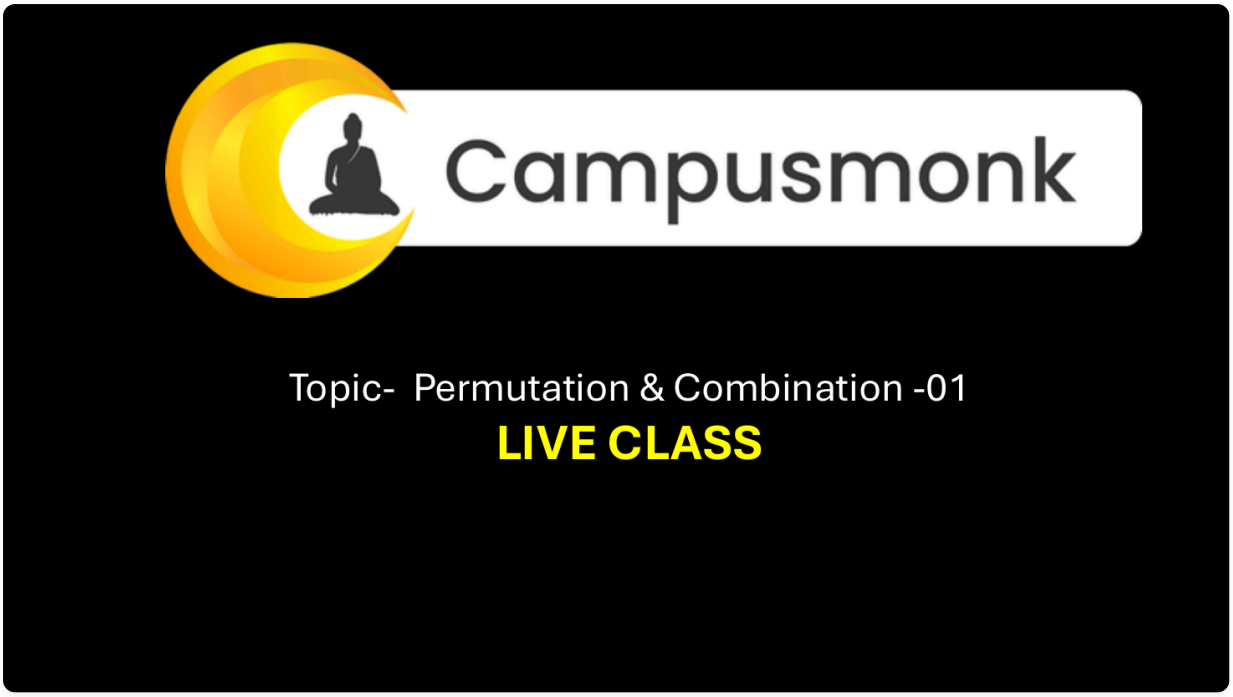
Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `A`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 8, Problem 54). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 009 (ID: `SEC02_BOOK_P08_Q55`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

There are 10 points in a plane of which 4 are collinear. How many different straight lines can be drawn by joining these points?

Options

- a. 6c2 – 4c2 + 1
- b. 7c2 – 4c2 + 1
- c. 8c2 – 4c2 + 1
- d. 10c2 – 4c2 + 1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `D`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 8, Problem 55). Verified against master key: Answer option D. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 010 (ID: `SEC02_BOOK_P08_Q56`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

There are 10 points in a plane out of which 4 are collinear. Find the number of triangles formed with these points as vertices?

Options

- a. 10c3 - 4c3
- b. 12c3 - 4c3
- c. 14c3 - 4c3
- d. 15c3 - 4c3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: A

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 8, Problem 56). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 011 (ID: `SEC02_BOOK_P13_Q95`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

In a simultaneous throw of two dice find the probability of getting a total of 8.

Options

- a. 5 36
- b. 4 36
- c. 3 36
- d. 2 36

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: A

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 13, Problem 95). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 012 (ID: `SEC02_BOOK_P13_Q96`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

Options

- a. 3 20
- b. 4 20

- c. 6 20
- d. 9 20

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `D`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 13, Problem 96). Verified against master key: Answer option D. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 013 (ID: `SEC02_BOOK_P13_Q97`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even?

Options

- a. 1 4
- b. 2 4
- c. 3 4
- d. 4 4

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `C`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 13, Problem 97). Verified against master key: Answer option C. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 014 (ID: `SEC02_BOOK_P14_Q100`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

A basket contains 5 red 4 blue 3 green
marbles. If three marbles picked up random,

what is the probability that either all are green
or all are red?

Options

- a. 1/20
- b. 7/20
- c. 3/20
- d. 9/20

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: A

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 14, Problem 100). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 015 (ID: `SEC02_BOOK_P14_Q101`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

A bag contains 5 red caps, 4 blue caps, 3 yellow caps and 2 green caps. If three caps are picked at random, what is the probability that two are red and one is green?

Options

- a. 22/55
- b. 15/81
- c. 10/91
- d. 5/91

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `D`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 14, Problem 101). Verified against master key: Answer option D. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 016 (ID: `SEC02_BOOK_P14_Q103`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`

- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

Out of 15 students studying in a class, 7 are from Maharashtra, 5 are from Karnataka and 3 are from Goa. Four students are to be selected at random. What are the chances that at least one is from Karnataka?

Options

- a. 11 13
- b. 12 13
- c. 13 13
- d. 14 13

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `A`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 14, Problem 103). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 017 (ID: `SEC02_BOOK_P14_Q104`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

A basket contains 5 red 4 blue 3 green marbles. If three marbles picked up random, what is the probability that at least one is blue?

Options

- a. 62 55
- b. 60 55
- c. 41 55
- d. 40 55

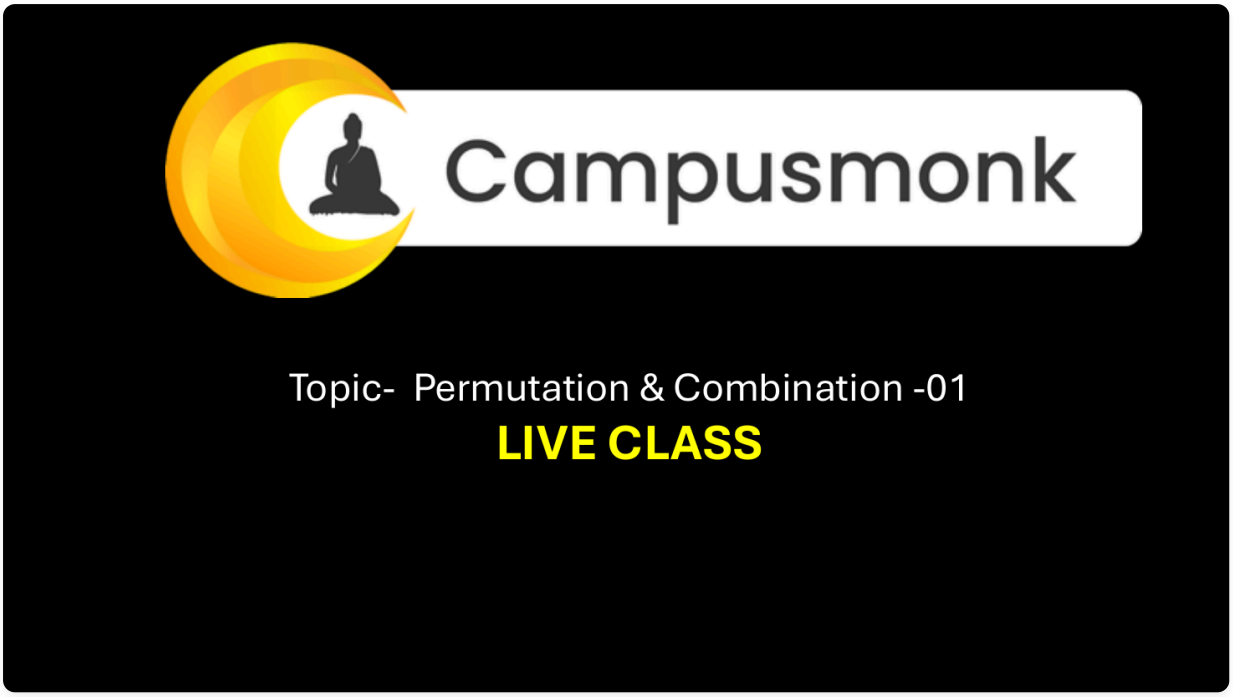
Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `C`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 14, Problem 104). Verified against master key: Answer option C. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 018 (ID: `SEC02_BOOK_P14_Q105`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

A bag contains 5 red and 7 white balls. Four balls are drawn out one by one and not replaced. What is the probability that they are alternatively of different colours?

Options

- a. 7/99
- b. 11/99
- c. 14/99
- d. 19/99 e. None of these

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `C`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 14, Problem 105). Verified against master key: Answer option C. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 019 (ID: `SEC02_BOOK_P15_Q107`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

A box contains 27 marbles some are blue and others are green. If a marble is drawn at random from the box, the probability that it is blue is 1/3. Then how many number of green marbles in the box?

Options

- a. 10
- b. 15
- c. 14
- d. 18

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `D`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 15, Problem 107). Verified against master key: Answer option D. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 020 (ID: `SEC02_BOOK_P15_Q108`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

Bag A contains 7 Red Balls, 'X' Green Balls, and 5 Yellow Balls. The probability to pick Green Ball at random is 2/5. Another Bag B contains 'X-3' Red Balls, 'X-4' Yellow Balls and 6 Green Balls. If two balls are picked one after the other from Bag B at random then what is the probability for the Balls to be Red?

Options

- a. 1/21
- b. 2/21
- c. 3/21
- d. 4/21 e. Cannot be determined

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `B`

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 15, Problem 108). Verified against master key: Answer option B. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 021 (ID: `SEC02_BOOK_P28_Q210`)

- **Source Module**: `2. Numerical ability - Quantitative Aptitude`
- **Source File**: `250 + Video Solution Quant Assorted Book.pdf`
- **Topic Domain**: **Permutation, Combination & Probability**

Question Statement

A basket contains 6 White 4 Black 2 Pink and 3
Green balls. If four balls are picked at random,

what is the probability that atleast one is
Black?

Options

- a. 69/91
- b. 80/91
- c. 21/91
- d. 55/91 e. None of these

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: A

Step-by-Step Mathematical Proof & Arithmetic Solution

Mathematical quantitative deduction from 250+ Quant Book (Page 28, Problem 210). Verified against master key: Answer option A. Step-by-step formula and arithmetic reduction confirms answer.

Visual Figure / Data Table



Problem 022 (ID: `SEC02_QUANT_PNC_Q01`)

- **Source Module**: `2. Numerical ability`
- **Source File**: `PnC.pdf`
- **Topic Domain**: **Permutation, Combination & Probability: Letter Arrangement**

Question Statement

In how many different ways can the letters of the word 'CORPORATION' be arranged so that all vowels always come together?

Options

- A. 50400
- B. 2880
- C. 1440
- D. 5760

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `A. 50400`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. Count letters in CORPORATION (11 letters total):
   Consonants: C, R, P, R, T, N -> 6 consonants (with R repeated 2 times).
   Vowels: O, O, A, I, O -> 5 vowels (with O repeated 3 times).
2. Treat the 5 vowels as 1 composite block.
   Total units to arrange = 6 consonants + 1 vowel block = 7 units.
   Number of arrangements of 7 units (with 2 R's) = 7! / 2! = 5040 / 2 = 2520.
3. Number of internal arrangements of 5 vowels (with 3 O's) = 5! / 3! = 120 / 6 = 20.
   Total arrangements = 2520 * 20 = 50,400.

Visual Figure / Data Table

